

DATA STRUCTURES

HOMEWORK-1

1. Sum of two matrices

Code:

```
#include <stdio.h>
int main() {
    int arr[50][50], arr1[50][50], sum[50][50];
    int i, j, k, l, a, b;
    printf("Enter the rows and columns of 1st matrix:");
    scanf("%d %d", &a, &b);
    printf("Enter the rows and columns of 2nd matrix:");
    scanf("%d %d", &k, &l);
    if (a != k || b != l) {
        printf("Matrix addition is not possible");
        return 0;
    }
    printf("Enter the values for matrix 1:");
    for (i = 0; i < a; i++) {
        for (j = 0; j < b; j++) {
            scanf("%d", &arr[i][j]);
        }
    }
    printf("Enter the values for matrix 2:");
    for (i = 0; i < a; i++) {
        for (j = 0; j < b; j++) {
            scanf("%d", &arr1[i][j]);
        }
    }
    for (i = 0; i < a; i++) {
        for (j = 0; j < b; j++) {
            sum[i][j] = arr[i][j] + arr1[i][j];
        }
    }
    printf("Added matrix:\n");
    for (i = 0; i < a; i++) {
        for (j = 0; j < b; j++) {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
● PS D:\ds_hw> gcc sum.c -o sum
● PS D:\ds_hw> ./sum
Enter the rows and columns of 1st matrix:2
2
Enter the rows and columns of 2nd matrix:2
2
Enter the values for matrix 1:10
20
30
40
Enter the values for matrix 2:21
32
43
34
Added matrix:
31 52
73 74
○ PS D:\ds_hw> █
```

2. Write a program in C to read n number of values in an array and display them in reverse order.

Code:

```
#include <stdio.h>
int main(){
    int arr[50],n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the values:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("Reverse order:\n");
    for(i=n-1;i>=0;i--){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw> gcc rev.c -o rev
PS D:\ds_hw> ./rev
Enter the number of elements:4
Enter the values:23
12
34
45
Reverse order:
45 34 12 23
PS D:\ds_hw>
```

3. Count the total number of duplicate elements in an array.

Code:

```
#include <stdio.h>
int main() {
    int arr[50], n, i, j, count=0, flag;
    printf("Enter the number of elements:");
    scanf("%d", &n);
    printf("Enter the elements:\n");
    for(i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }
    for(i=0; i<n; i++) {
        flag=0;
        for(j=0; j<i; j++) {
            if(arr[i]==arr[j]) {
                flag=1;
                break;
            }
        }
        if(flag)
            continue;

        for(j=i+1; j<n; j++) {
            if(arr[i]==arr[j]) {
                count++;
            }
        }
    }
    printf("\nThe number of duplicates are %d", count);
    return 0;
}
```

Output:

```
43 34 12 23
● PS D:\ds_hw> gcc dup.c -o dup
● PS D:\ds_hw> ./dup
Enter the number of elements:5
Enter the elements:
12
32
1
234
32

The number of duplicates are 1
○ PS D:\ds_hw> █
```

4. Print unique element in an array.

Code:

```
#include <stdio.h>
int main(){
    int arr[50],n,i,j,value=0,flag;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the values:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("The unique elements are:\n");
    for(i=0;i<n;i++){
        value=0;
        flag=0;
        for(j=0;j<i;j++){
            if(arr[i]==arr[j]){
                flag=1;
                break;
            }
            if(flag)
                continue;
        }
        for(j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                value=1;
            }
        }

        if(value==0){
            printf("%d ",arr[i]);
        }
    }
    return 0;
}
```

Output:

```
● PS D:\ds_hw> ./uni
Enter the number of elements:5
Enter the values:12
23
12
12
34
The unique elements are:
23 34
○ PS D:\ds_hw> █
```

5. Insert in sorted array.

Code:

```
#include <stdio.h>
int main(){
int arr[20],n,a,i,pos=0;
printf("Enter the number of elements:");
scanf("%d",&n);
printf("Enter the sorted array:");
for(i=0;i<n;i++){
scanf("%d",&arr[i]);
}
printf("Enter the value to be inserted:");
scanf("%d",&a);
for(i=0;i<n;i++){
if(arr[i]==a){
pos=i;
break;
}
if(i==n-1)
continue;
else if(arr[i]<a && arr[i+1]>a){
pos=i;
break;
}
}
if(arr[0]>a)
pos=0;
else
if(arr[n-1]<a)
pos=n-1;
if(a<arr[0]){
for(i=n;i>0;i--){
arr[i]=arr[i-1];
}
arr[pos]=a;
}
```

```

}
else{
for(i=n;i>pos+1;i--){
    arr[i]=arr[i-1];
}
arr[pos+1]=a;}
n++;
printf("The inserted array is:");
for(i=0;i<n;i++){
    printf("%d ",arr[i]);
}
return 0;
}

```

Output:

```

● PS D:\ds_hw> ./sort
Enter the number of elements:4
Enter the sorted array:12
16
34
56
Enter the value to be inserted:45
The inserted array is:12 16 34 45 56
○ PS D:\ds_hw>

```

6. Insert in unsorted array.

Code:

```

#include <stdio.h>

int main(){
    int arr[50],n,a,p,i;

    printf("Enter the number of elements:");
    scanf("%d",&n);

    printf("Enter the elements:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }

    printf("Enter the position and the element to be inserted:");
    scanf("%d %d",&p,&a);

    if(p<1 || p>n+1){
        printf("Invalid position");
    }
}

```

```

return 0;
}

for(i=n;i>=p;i--){
    arr[i]=arr[i-1];
}

arr[p-1]=a;
n++;

printf("Inserted array is:");
for(i=0;i<n;i++){
    printf("%d ",arr[i]);
}

return 0;
}

```

Output:

```

PS D:\ds_hw> gcc unsort.c -o unsort
PS D:\ds_hw> ./unsort
Enter the number of elements:5
Enter the elements:12
98
67
34
234
Enter the position and the element to be inserted:2
100
Inserted array is:12 100 98 67 34 234
PS D:\ds_hw>

```

7. Matrix multiplication.

Code:

```

#include <stdio.h>

int main(){
    int arr[50][50],arr1[50][50],multi[50][50],i,j,k,r1,r2,c1,c2;
    printf("Enter the rows and the columns of the matrix 1:");
    scanf("%d %d",&r1,&c1);
    printf("Enter the rows and the columns of the matrix 2:");
    scanf("%d %d",&r2,&c2);
    if(c1!=r2){
        printf("Matrix multiplication is not possible");
        return 0;
    }
    printf("Enter the values for matrix 1:\n");
    for(i=0;i<r1;i++){
        for(j=0;j<c1;j++){
            scanf("%d",&arr[i][j]);
        }
        printf("\n");
    }
}

```

```

}

printf("Enter the values for matrix 2:\n");
for(i=0;i<r2;i++){
    for(j=0;j<c2;j++){
        scanf("%d",&arr1[i][j]);
    }
    printf("\n");
}
for(i=0;i<r1;i++){
    for(j=0;j<c2;j++){
        multi[i][j]=0;
        for(k=0;k<c1;k++){
            multi[i][j]+=arr[i][k]*arr1[k][j];
        }
    }
}
for(i=0;i<r1;i++){
    for(j=0;j<c2;j++){
        printf("%d ",multi[i][j]);
    }
    printf("\n");
}
return 0;
}

```

Output:

```

● PS D:\ds_hw> ./multi
Enter the rows and the columns of the matrix 1:2
2
Enter the rows and the columns of the matrix 2:2
2
Enter the values for matrix 1:
1
2

3
4

Enter the values for matrix 2:
5
6

7
8

19 22
43 50
○ PS D:\ds_hw>

```


8. Matrix Transpose

Code:

```
#include <stdio.h>

int main(){
    int arr[50][50],trans[50][50],i,j,r,c;
    printf("Enter the number of rows and columns:");
    scanf("%d %d",&r,&c);
    printf("Enter the values:");
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            trans[j][i]=arr[i][j];
        }
    }
    for(i=0;i<c;i++){
        for(j=0;j<r;j++){
            printf("%d ",trans[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw> gcc transpose.c -o transpose
PS D:\ds_hw> ./transpose
Enter the number of rows and columns:3
2
Enter the values:22
34
12
56
78
56
22 12 78
34 56 56
PS D:\ds_hw> 
```

9. Right diagonal sum

Code:

```
#include <stdio.h>

int main(){
    int arr[50][50],i,j,sum=0,r,c;
    printf("Enter the row and columns:");
    scanf("%d %d",&r,&c);
    if(r!=c){
        printf("Diagonal addition is not possible");
        return 0;
    }
    printf("Enter the elements:");
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            if(i==j){
                sum+=arr[i][j];
            }
        }
    }
    printf("Sum of the diagonal:%d",sum);
    return 0;
}
```

Output:

```
PS D:\ds_hw> ./right
Enter the row and columns:3
3
Enter the elements:23
34
54
12
78
90
78
23
11
Sum of the diagonal:112
PS D:\ds_hw>
```

10. Check for identity matrix

Code:

```
#include <stdio.h>

int main(){
    int arr[50][50],i,j,r,c,value=0;
    printf("Enter the number of rows and columns:");
    scanf("%d %d",&r,&c);
    if(r!=c){
        printf("Invalid input");
        return 0;
    }
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    for(i=0;i<r;i++){
        for(j=0;j<c;j++){
            if((i==j && arr[i][j]!=1) || (i!=j && arr[i][j]!=0)){
                value=1;
            }
        }
    }
    if(value!=0){
        printf("Given matrix is not identity");
    }
    else{
        printf("\nIt's an identity matrix");
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw> gcc identity.c -o identity
PS D:\ds_hw> ./identity
Enter the number of rows and columns:2
2
1
0
0
1

It's an identity matrix
PS D:\ds_hw>
```

11. To find number of vowels, consonants ,digits and white space.

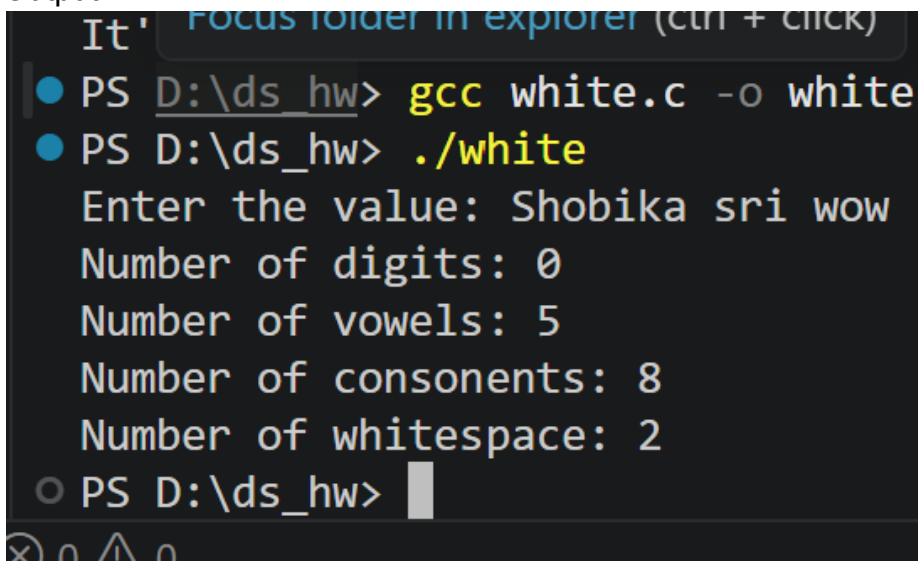
Code:

```
#include <stdio.h>

int main(){
    char arr[100];
    int i,digit=0,vowel=0,consonent=0,white=0;
    printf("Enter the value: ");
    fgets(arr,100,stdin);
    for(i=0;arr[i]!='\0';i++){
        if(arr[i]>=48 && arr[i]<=57){
            digit++;
        }
        if(arr[i]>=65 && arr[i]<=90 || arr[i]>=97 && arr[i]<=122){
            if(arr[i]=='A' ||arr[i]=='a' ||arr[i]=='E' ||arr[i]=='e' ||arr[i]=='i'
||arr[i]=='I' ||arr[i]=='o' ||arr[i]=='O' ||arr[i]=='u' ||arr[i]=='U'){
                vowel++;
            }

            else{
                consonent++;
            }
        }
        if(arr[i]==' '){
            white++;
        }
    }
    printf("Number of digits: %d",digit);
    printf("\nNumber of vowels: %d",vowel);
    printf("\nNumber of consonents: %d",consonent);
    printf("\nNumber of whitespace: %d",white);
}
```

Output:



```
It's Focus folder in explorer (ctrl + click)
● PS D:\ds_hw> gcc white.c -o white
● PS D:\ds_hw> ./white
Enter the value: Shobika sri wow
Number of digits: 0
Number of vowels: 5
Number of consonents: 8
Number of whitespace: 2
○ PS D:\ds_hw>
```

12. Frequency of a character in a string.

Code:

```
#include <stdio.h>

int main(){
    char arr[100];
    int freq[256]={0};
    int i,j,value=0;
    printf("Enter the sentence:");
    fgets(arr,100,stdin);
    for(i=0;arr[i]!='\0';i++){
        freq[arr[i]]++;
    }
    printf("The frequency of the characters:\n");
    for(i=0;arr[i]!='\0';i++){
        value=0;
        for(j=0;j<i;j++){
            if(arr[i]==arr[j])
                value=1;
        }
        if(value){
            continue;
        }
        if(arr[i]>='a' && arr[i]<='z' || arr[i]>='A' && arr[i]<='Z'){
            if(freq[arr[i]]>=1){
                printf("%c - Frequency %d\n",arr[i],freq[arr[i]]);
            }
        }
    }
    return 0;
}
```

Output:

```
● PS D:\ds_hw> gcc freq.c -o freq
● PS D:\ds_hw> ./freq
Enter the sentence:shobika sri
The frequency of the characters:
s - Frequency 2
h - Frequency 1
o - Frequency 1
b - Frequency 1
i - Frequency 2
k - Frequency 1
a - Frequency 1
r - Frequency 1
● PS D:\ds_hw> 
```

13. Count words in a string.

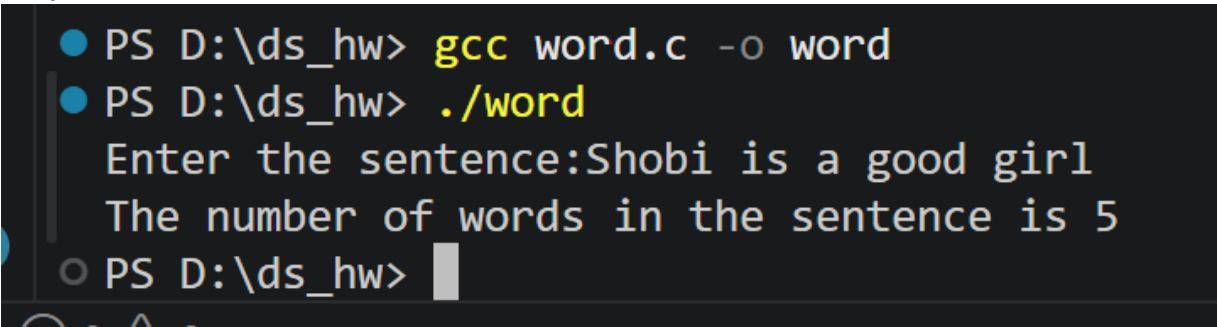
Code:

```
#include <stdio.h>

int main(){
    char arr[100];
    int i, words=0, d=0;
    printf("Enter the sentence:");
    fgets(arr, 100, stdin);
    for(i=0; arr[i]!='\0'; i++){
        if(arr[i] == ' ' && arr[i+1]!=' '){
            continue;
        }
        if(arr[i]!=0 && d==0){
            words++;
            d=1;
        }
        else if(arr[i]==' '){
            d=0;
        }
    }

    printf("The number of words in the sentence is %d", words);
    return 0;
}
```

Output:



```
PS D:\ds_hw> gcc word.c -o word
PS D:\ds_hw> ./word
Enter the sentence:Shobi is a good girl
The number of words in the sentence is 5
PS D:\ds_hw>
```

14. Sort string array

Code:

```
#include <stdio.h>
#include <string.h>

int main(){
    char arr[50][50], temp[50];
    int i, n, j;
    printf("Enter the number of words:");
    scanf("%d", &n);
    printf("Enter the words:");
    for(i=0; i<n; i++){
        scanf("%s", arr[i]);
    }
}
```

```

    }

    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if((strcmp(arr[i],arr[j]))>0){
                strcpy(temp,arr[i]);
                strcpy(arr[i],arr[j]);
                strcpy(arr[j],temp);
            }
        }
    }

    printf("Sorted array:\n");
    for(i=0;i<n;i++){
        printf("%s\n",arr[i]);
    }

    return 0;
}

```

Output:

```

PS D:\ds_hw> gcc sorts.c -o sorts
PS D:\ds_hw> ./sorts
Enter the number of words:4
Enter the words:shobi
sangee
murugan
nagul
Sorted array:
murugan
nagul
sangee
shobi
PS D:\ds_hw>

```

15.Toggle case of a sentence

Case:

```

#include <stdio.h>
int main(){
    char arr[50];
    int i,n;
    printf("Enter the sentence:");
    fgets(arr,50,stdin);
    printf("Enter whether to convert from : \n1.Lower to upper\n2.Upper to Lower\n");
    scanf("%d",&n);
    switch(n){
        case 1:
            for(i=0;arr[i]!='\0';i++){
                if(arr[i]>='a' && arr[i]<='z'){

```

```

        arr[i]=arr[i]-32;
    }
}
break;
case 2:
for(i=0;arr[i]!='\0';i++){
    if(arr[i]>='A' && arr[i]<='Z'){
        arr[i]=arr[i]+32;
    }
}
break;
default:
    printf("Invalid input");
    break;
}
printf("Converted string:\n");
for(i=0;arr[i]!='\0';i++){
    printf("%c",arr[i]);
}
return 0;
}

```

Output:

```

● PS D:\ds_hw> gcc toggle.c -o toggle
● PS D:\ds_hw> ./toggle
Enter the sentence:Mind is everything,what you think you become.
Enter whether to convert from :
1.Lower to upper
2.Upper to Lower
1
Converted string:
MIND IS EVERYTHING,WHAT YOU THINK YOU BECOME.
● PS D:\ds_hw>

```